Yichen Wang, M.Sc. in Computer Science

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SUMMARY

Software Engineer with 6+ years of experience building scalable, cloud-native web applications and APIs in production environments. Strong background in **JavaScript/TypeScript, React, Node.js, and AWS** infrastructure. Experienced in shipping reliable systems, improving test coverage, optimizing performance, and delivering features end-to-end from development through production deployment.

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, SQL
- **Frontend:** React.js, Next.js, Svelte, React Native, Tailwind CSS, Responsive Design
- **Backend:** Node.js (Express), Flask, RESTful APIs
- **Cloud & Infra:** AWS (EC2, S3, Lambda, DynamoDB, API Gateway), Docker
- **Databases:** PostgreSQL, DynamoDB
- **Testing:** Jest, Playwright, CI/CD (GitHub Actions)

PROFESSIONAL EXPERIENCE

IBM

Senior Software Engineer *02/2022 - PRESENT*

- Delivered 500+ production enhancements across three major releases of a high-traffic public-facing enterprise application.
- Led frontend modernization initiative (React → Svelte), improving maintainability and rendering performance.
- Increased automated test coverage from 40% to 85% by authoring 300+ tests (Jest + Playwright), reducing production incidents by 60%.
- Coordinated DEV → UAT → PROD release cycles, ensuring stable deployments and fast incident resolution.
- Designed multilingual content and automation pipelines using Strapi and AWS services to improve global accessibility.
- Collaborated closely with backend engineers on API integrations, debugging, and performance optimization.

University at Buffalo

Full Stack Developer *11/2019 - 07/2021*

- Built and maintained React-based research platforms with dynamic dashboards, forms, and structured data workflows.
- Developed backend utilities (PHP + CSV pipelines) for data export and processing.
- Improved UI accessibility and responsive design across research applications.
- Worked closely with researchers in an Agile, iterative delivery environment.

INDEPENDENT PROJECT EXPERIENCE

Cross-Platform Social App (React Native + AWS + AI integrations)

- Architected and built a scalable social platform using **React/React Native, Node.js, PostgreSQL, and AWS.**
- Designed **REST APIs** supporting posts, threaded discussions, authentication, and content moderation.
- Implemented asynchronous multilingual pipelines using AWS Translate.
- Integrated AI-powered summarization and sentiment analysis via **OpenAI APIs**.
- Deployed backend services on **AWS EC2 with S3** storage and automated moderation checks (**AWS Rekognition**).

Lost & Found Pet Platform (React + Supabase + Map integrations)

- Built a geospatial React application using **Supabase (PostgreSQL + Auth)**.
- Implemented real-time updates, map-based discovery, and structured reporting workflows.
- Designed scalable data models and optimized query performance.

EDUCATION

University at Buffalo — Buffalo, NY

- M.S. in Computer Science and Engineering — May 2020
- B.S. in Computer Science, Magna Cum Laude — Dec 2018